Averaging Nonlinear Lipschitz Extension Operators

Automated discovery packet

June 15, 2026

Source Question

In Remark 2.2(ii) of arXiv:2001.09303, Sofi asks whether the Hilbert-space conclusion in Corollary 2.4 remains valid if the extension operator

$$F : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$$

is assumed to be Lipschitz rather than linear, for each subspace Z of the Banach space X. The question appears on page 10:

[21], Chapter 6.2.3). □

***Remark* 2.2.** (i) In contrast to Theorems 2.3 and 2.10 involving the existence /nonexistence of extension operators from spaces of Lipschitz functions on (closed) subspaces and arbitrary subsets of Banach spaces respectively, Kopecka [16] shows that in the latter case involving arbitrary subsets S of a Hilbert space H, there exist ‘extension’ operators $\psi : \mathrm{Lip}(S,H) \to \mathrm{Lip}(B,H)$ preserving the Lipschitz norm and acting between spaces of vector-valued bounded Lipschitz functions which are continuous but not linear. Here the spaces of Lipschitz functions are equipped with the sup-norm.

(ii) In light of Theorem 2.3 and the corollaries following from it, a natural question arises regarding the extent of the validity of these conclusions under the assumption that for a subspace Z of X, the extension operator $F : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$ is assumed to be Lipschitz rather than linear. To the best of our understanding, the question does not seem to have been addressed in the literature so far. Accordingly, we pose it as an open problem:

Question: Let X be a Banach space. Does the existence of a Lipschitz extension operator $F : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$ for each subspace Z of X yield that X is a Hilbert space?

We note that for a subspace Z of X for which $\mathrm{Lip}_0(Z)$ is separable, the answer is in the affirmative. This follows from an important theorem of Godefroy and Kalton [10] to the effect that given Banach spaces X and Z with Z separable and a bounded linear operator $T : X \longrightarrow Z$, the existence of a Lipschitz right inverse $S : Z \longrightarrow X$ of T yields the existence of a bounded linear right inverse $L : Z \longrightarrow X$ of $T : T \circ L = Id_Z$. Indeed, consider the bounded linear map $\psi : \mathrm{Lip}_0(X) \longrightarrow \mathrm{Lip}_0(Z)$ given by: $\psi(f) = f_{|Z}$. Now the presumed existence of $F : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$ as a Lipschitz extension operator gives that F is a Lipschitz right inverse of ψ. By the Godefroy-Kalton theorem just stated, we get a bounded linear right inverse G of ψ, i.e., $G : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$ is a bounded linear map such that $\psi \circ G = Id_{Lip0(Z)}$. In other words, G is a bounded linear extension operator and so we are in the situation of Theorem 2.3 to arrive at the desired conclusion.

We conclude with a brief discussion of how, on the one hand, the parameter

Result

The answer is affirmative. In fact, it is enough to assume the existence of such a Lipschitz extension operator for every closed subspace Z of X.

Proof Intuition

The restriction map $\rho_Z : \mathrm{Lip}_0(X) \to \mathrm{Lip}_0(Z)$ is a bounded linear surjection. A Lipschitz extension operator is exactly a nonlinear Lipschitz right inverse of this map. Since $\mathrm{Lip}_0(X)$ is canonically the dual of the Lipschitz-free space $\mathcal{F}(X)$, one can average the difference translates

$$F(g+h) - F(g), \qquad g \in \mathrm{Lip}_0(Z),$$

using an invariant mean on the additive abelian group $\mathrm{Lip}_0(Z)$. Translation invariance forces additivity in h, and weak-star compactness keeps the average inside $\mathrm{Lip}_0(X)$. This converts the nonlinear right inverse into a bounded linear right inverse, after which Sofi’s Corollary 2.4 applies.

A Linearization Lemma

Lemma 1. *Let X be a Banach space and let $Z \subset X$ be a closed subspace. Suppose that there is a Lipschitz map*

$$F : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$$

such that $F(f)|_Z = f$ for every $f \in \mathrm{Lip}_0(Z)$. Then there is a bounded linear map

$$G : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$$

such that $G(f)|_Z = f$ for every $f \in \mathrm{Lip}_0(Z)$. Moreover one may take $\|G\| \le \mathrm{Lip}(F)$ after replacing F by $F - F(0)$.

Proof. Let $\rho : \mathrm{Lip}_0(X) \to \mathrm{Lip}_0(Z)$ be the restriction map. Since $\rho F = \mathrm{Id}_{\mathrm{Lip}_0(Z)}$, we have $\rho F(0) = 0$. Replacing F by $F - F(0)$, we may assume $F(0) = 0$ and still have $\rho F = \mathrm{Id}$. Put $Y = \mathrm{Lip}_0(Z)$, $V = \mathrm{Lip}_0(X)$, and let $L = \mathrm{Lip}(F)$.

We use the standard invariant mean on the additive abelian group Y. Thus there is a positive norm-one functional m on $\ell_\infty(Y)$ such that

$$m(\varphi(\,\cdot + a\,)) = m(\varphi) \qquad (a \in Y,\ \varphi \in \ell_\infty(Y)).$$

Recall that $V = \mathcal{F}(X)^*$, where $\mathcal{F}(X)$ is the Lipschitz-free space over X. For $h \in Y$, define a bounded V-valued function on Y by

$$\Delta_h(g) = F(g + h) - F(g), \qquad g \in Y.$$

It is bounded because

$$\|\Delta_h(g)\|_{\mathrm{Lip}} \le L\|h\|_{\mathrm{Lip}} \qquad (g \in Y).$$

Define $Gh \in \mathcal{F}(X)^* = V$ by weak-star averaging:

$$\langle \mu, Gh\rangle = m\big(g \mapsto \langle \mu, \Delta_h(g)\rangle\big), \qquad \mu \in \mathcal{F}(X).$$

This gives $\|Gh\|_{\mathrm{Lip}} \le L\|h\|_{\mathrm{Lip}}$.

We first check linearity. For $h, k \in Y$,

$$\Delta_{h+k}(g) = F(g + h + k) - F(g) = \Delta_k(g + h) + \Delta_h(g).$$

Translation invariance of m therefore gives

$$G(h + k) = Gk + Gh.$$

Thus G is additive. Since G is Lipschitz at the origin, additivity implies real linearity.

It remains to verify that G is still an extension operator. Let $J : \mathcal{F}(Z) \to \mathcal{F}(X)$ be the canonical isometric linear embedding induced by the inclusion $Z \subset X$. Then $\rho = J^*$. For $\nu \in \mathcal{F}(Z)$ and $h \in Y$,

$$\begin{aligned}
\langle \nu, \rho G h\rangle &= \langle J\nu, Gh\rangle \\
&= m\big(g \mapsto \langle J\nu, F(g + h) - F(g)\rangle\big) \\
&= m\big(g \mapsto \langle \nu, \rho F(g + h) - \rho F(g)\rangle\big) \\
&= m\big(g \mapsto \langle \nu, (g + h) - g\rangle\big) \\
&= \langle \nu, h\rangle.
\end{aligned}$$

Hence $\rho G h = h$ for every $h \in \mathrm{Lip}_0(Z)$, i.e. $G(h)|_Z = h$. This is the desired bounded linear extension operator. $\square$

Solution of Sofi's Question

Theorem 1. *Let X be a Banach space. Assume that for every closed subspace $Z \subset X$ there exists a Lipschitz extension operator*

$$F_Z : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X).$$

Then X is isomorphic to a Hilbert space. Consequently the question in Remark 2.2(ii) has an affirmative answer.

Proof. By the lemma, for every closed subspace $Z \subset X$, the presumed Lipschitz extension operator F_Z yields a bounded linear extension operator

$$G_Z : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X).$$

Sofi's Corollary 2.4 states that, for a Banach space X, the existence of a bounded linear extension operator $\mathrm{Lip}_0(Z) \to \mathrm{Lip}_0(X)$ for every closed subspace $Z \subset X$ is equivalent to X being isomorphic to a Hilbert space. Applying that corollary gives the conclusion. □

Corollary 1. *For a Banach space X, the following are equivalent:*

1. *X is isomorphic to a Hilbert space.*
2. *For every closed subspace $Z \subset X$, there exists a bounded linear extension operator $\mathrm{Lip}_0(Z) \to \mathrm{Lip}_0(X)$.*
3. *For every closed subspace $Z \subset X$, there exists a Lipschitz extension operator $\mathrm{Lip}_0(Z) \to \mathrm{Lip}_0(X)$.*

Proof. The equivalence of (1) and (2) is Sofi's Corollary 2.4. The implication (2)⇒(3) is immediate, since bounded linear maps are Lipschitz. The implication (3)⇒(2) is the linearization lemma applied to each closed subspace Z. □

Verification Notes

The proof is deterministic and uses only the standard invariant mean on an abelian group and the canonical duality $\mathrm{Lip}_0(M) = \mathcal{F}(M)^*$. The critical identity is $\rho = J^*$, where $J : \mathcal{F}(Z) \to \mathcal{F}(X)$ is induced by the inclusion $Z \subset X$. This makes the weak-star average compatible with restriction and proves $\rho G = \mathrm{Id}_{\mathrm{Lip}_0(Z)}$.

The argument does not address the separate questions in the source about Lipschitz retracts of biduals, type 2, or the parameter $\lambda(M)$.

Novelty and Literature Check

Before promotion, the local registry, solution, attempt, and proof-gap indexes were searched for arXiv:2001.09303 and for the core phrases $\mathrm{Lip}_0(Z) \to \mathrm{Lip}_0(X)$, "Lipschitz extension operator", "linear right inverse", and "invariant mean". No existing run packet for this question was found.

Web searches on June 15, 2026 for the exact question and for variants of "Lipschitz right inverse linear right inverse invariant mean Banach" found the source arXiv paper and related invariant-mean linearization material, but no explicit later answer to this $\mathrm{Lip}_0(Z) \to \mathrm{Lip}_0(X)$ question. The supporting Baudier–Lancien manuscript records the classical invariant-mean linearization background, including vector-valued invariant means and Lindenstrauss' linear extension theorem, but the exact application above is the packet's contribution subject to human review.

References

[1] M. A. Sofi, *Extension operators and nonlinear structure of Banach spaces*, arXiv:2001.09303, 2020.

[2] F. Baudier and G. Lancien, *Nonlinear geometry of Banach spaces*, arXiv:2512.00817, 2025.

[3] G. Godefroy and N. J. Kalton, *Lipschitz-free Banach spaces*, Studia Mathematica 159 (2003), 121–141.

[4] J. Lindenstrauss, *On nonlinear projections in Banach spaces*, Michigan Mathematical Journal 11 (1964), 263–287.

Human Review: Linearizing Lipschitz Extension Operators

Original automatic proof: `solution_packet.pdf`
Checked date: 15 June 2026
Status: Manually checked.
Verdict: The proof appears correct.

Human Comments

The proof appears to correctly answer the question from Sofi's Remark 2.2(ii). The key point is that a Lipschitz, not necessarily linear, extension operator

$$F : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X), \qquad F(f)|_Z = f,$$

can be replaced by a bounded linear extension operator by averaging its finite differences over the additive abelian group $\mathrm{Lip}_0(Z)$. Once this linearization is obtained, Sofi's Corollary 2.4 applies and gives that X is isomorphic to a Hilbert space.

The idea is sound. However, the notation in the automatic proof is slightly more abstract than necessary. Introducing temporary symbols such as Y and V makes the proof harder to read, especially at the averaging step. It is clearer to keep the spaces visible throughout: the domain group is $\mathrm{Lip}_0(Z)$, and the values lie in $\mathrm{Lip}_0(X) = \mathcal{F}(X)^*$.

Notation-Improved Verification of the Averaging Step

Assume, after replacing F by $F - F(0)$, that

$$F(0) = 0, \qquad F(f)|_Z = f$$

for every $f \in \mathrm{Lip}_0(Z)$. Let

$$L = \mathrm{Lip}(F).$$

Since $\mathrm{Lip}_0(Z)$ is an additive abelian group, it admits an invariant mean. Thus there is a positive norm-one functional

$$m : \ell_\infty(\mathrm{Lip}_0(Z)) \longrightarrow \mathbb{R}$$

such that

$$m(\varphi(\cdot + a)) = m(\varphi) \qquad (a \in \mathrm{Lip}_0(Z),\ \varphi \in \ell_\infty(\mathrm{Lip}_0(Z))).$$

For each $h \in \mathrm{Lip}_0(Z)$, define the bounded $\mathrm{Lip}_0(X)$-valued function

$$\Delta_h : \mathrm{Lip}_0(Z) \longrightarrow \mathrm{Lip}_0(X)$$

by

$$\Delta_h(g) = F(g + h) - F(g), \qquad g \in \mathrm{Lip}_0(Z).$$

This is bounded because

$$\|\Delta_h(g)\|_{\mathrm{Lip}} = \|F(g + h) - F(g)\|_{\mathrm{Lip}} \le L\|h\|_{\mathrm{Lip}}$$

for every $g \in \mathrm{Lip}_0(Z)$.

Now use the canonical duality

$$\mathrm{Lip}_0(X) = \mathcal{F}(X)^*.$$

The invariant mean m only acts on scalar-valued bounded functions, so the average of Δ_h is defined weak-star coordinate by weak-star coordinate. Namely, define $G(h) \in \mathcal{F}(X)^* = \mathrm{Lip}_0(X)$ by

$$\langle \mu, G(h)\rangle = m\big(g \mapsto \langle \mu, \Delta_h(g)\rangle\big), \qquad \mu \in \mathcal{F}(X).$$

For fixed $\mu \in \mathcal{F}(X)$, the function

$$g \mapsto \langle \mu, \Delta_h(g)\rangle$$

is bounded, since

$$|\langle \mu, \Delta_h(g)\rangle| \le \|\mu\| \, \|\Delta_h(g)\|_{\mathrm{Lip}} \le L\|h\|_{\mathrm{Lip}}\|\mu\|.$$

Therefore the definition is legitimate. Moreover,

$$|\langle \mu, G(h)\rangle| \le L\|h\|_{\mathrm{Lip}}\|\mu\|,$$

and hence

$$\|G(h)\|_{\mathrm{Lip}} \le L\|h\|_{\mathrm{Lip}}.$$

Linearity follows from translation invariance of the mean. Indeed, for $h, k \in \mathrm{Lip}_0(Z)$,

$$\Delta_{h+k}(g) = F(g+h+k) - F(g) = \Delta_k(g+h) + \Delta_h(g).$$

Averaging over g, the first term satisfies

$$m\big(g \mapsto \langle \mu, \Delta_k(g+h)\rangle\big) = m\big(g \mapsto \langle \mu, \Delta_k(g)\rangle\big)$$

by invariance of m. Thus

$$G(h+k) = G(h) + G(k).$$

Since G is additive and satisfies

$$\|G(h)\|_{\mathrm{Lip}} \le L\|h\|_{\mathrm{Lip}},$$

it is continuous at the origin. Additivity gives $G(qh) = qG(h)$ for rational q, and continuity then extends this identity to every real scalar. Hence G is real linear.

Finally, G is still an extension operator. Let

$$\rho : \mathrm{Lip}_0(X) \longrightarrow \mathrm{Lip}_0(Z)$$

be restriction. If

$$J : \mathcal{F}(Z) \longrightarrow \mathcal{F}(X)$$

is the canonical map induced by the inclusion $Z \subset X$, then

$$\rho = J^*.$$

For $\nu \in \mathcal{F}(Z)$,

$$\begin{aligned}
\langle \nu, \rho G(h)\rangle &= \langle J\nu, G(h)\rangle \\
&= m\big(g \mapsto \langle J\nu, F(g+h) - F(g)\rangle\big) \\
&= m\big(g \mapsto \langle \nu, \rho F(g+h) - \rho F(g)\rangle\big) \\
&= m\big(g \mapsto \langle \nu, (g+h) - g\rangle\big) \\
&= \langle \nu, h\rangle.
\end{aligned}$$

Therefore

$$\rho G(h) = h,$$

so $G(h)|_Z = h$. This proves that the original Lipschitz extension operator can be linearized into a bounded linear extension operator.

Review Outcome

Archive this packet as an apparently correct full solution, pending any later author or literature feedback. The proof's central averaging argument looks correct; in a polished version, I would recommend using the notation above rather than the more compressed Y, V notation, since it makes clear that Δ_h is a bounded $\mathrm{Lip}_0(X)$-valued function on $\mathrm{Lip}_0(Z)$, and that the invariant mean is applied only after pairing with elements of $\mathcal{F}(X)$.